

Chapter 2: Professionalism and Codes of Ethics

HUM 4441: Engineering Ethics

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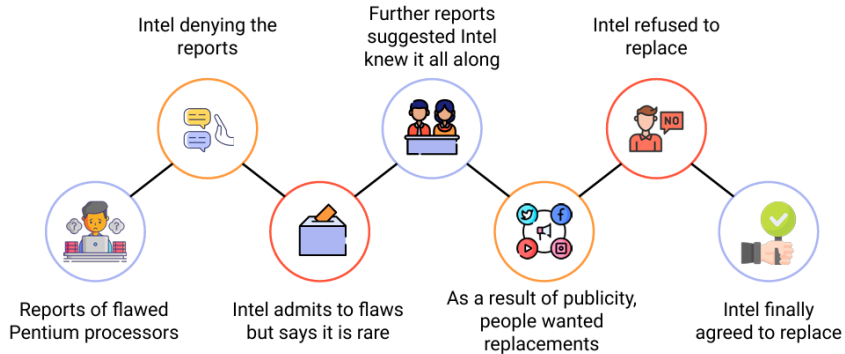
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Story of Intel



Was this ethical from Intel?

Chapter Objectives

After reading this chapter, you will be able to:

01

Determine whether engineering is a **profession**

02

Understand what **codes of ethics** are and how they work

03

Examine codes of ethics of **professional engineering societies**

Job, Occupation, or Profession?

Not all work qualifies as a profession

Job

Any work for hire regardless of skill or responsibility.

Engineering is a job - but much more.

Occupation

Employment through which one makes a living.

Engineering is an occupation - but still more.

Profession

Requires specific formal attributes beyond skill and pay.

Engineering meets all four criteria.

Note: “Professional athlete” and “professional carpenter” are **not** professions in this formal sense.

Four Attributes of a Profession

What makes something a true profession?

1. Skills, Judgment & Discretion

Work is not routine and cannot be mechanised.
Decisions have serious impacts on people's lives.

3. Special Governing Societies

Societies controlled by members set and enforce standards for admission and professional conduct.

2. Extensive Formal Education

Membership requires formal education; not simply practical training or apprenticeship.

4. Significant Public Good

Significant public benefit must result from the practice of the profession.

Understanding Judgment and Discretion

Judgment

Making **significant decisions** based on formal training and experience.

These decisions have serious impacts on people's lives and often involve large sums of money.

Not the everyday judgment of a secretary or mechanic.

Discretion - Two Meanings

1. Confidentiality: Keeping client and employer information private - essential for trust.

2. Autonomy: The ability to make important decisions independently.

Both separate a profession from a mere job.

High pay alone does **not** make someone a professional. Entertainers and athletes are highly paid - yet are not professionals.

Medicine and Law: True Professions

Two clear examples that meet every criterion

Medicine ✓

- Sophisticated skills; cannot be mechanised
- 8+ years of formal education
- Governed by the **AMA**
- Physician–patient confidentiality privilege
- Serves public good: healing the sick

Law ✓

- Sophisticated skills; extensive formal training
- Governed by the **ABA**
- Attorney–client privilege
- State licence required to practice
- Serves public good: access to justice

Is Engineering a Profession?

Checking engineering against all four criteria

Sophisticated skills & judgment

Design requires judgment on materials, trade-offs, and public safety



Formal education

4-year undergraduate degree minimum; many roles require advanced degrees



Professional societies

IEEE, ASME, NSPE exist - but weaker than AMA / ABA



Public good

Communications, transport, energy, medical technology



Verdict: YES - Engineering IS a profession.

Differences Between Engineering and Other Professions

These are aspects which are currently lacking in engineering

	Engineering	Others (Law / Medicine)
How practiced	Mostly as employees in large companies	Can also practice privately or independently
Training	Finishes within 4 years of undergraduate	Begins <i>after</i> 4 years of undergraduate
Licences	Most engineers do <i>not</i> hold engineering licences	Licences are compulsory in many professions
Societies	Many discipline-specific; relatively weak	Single unified body (AMA/ABA); very powerful

Two Models of Professional Societies

Why do professional societies exist?

Social-Contract Model

Societies exist primarily to **serve the public good**.

Society grants high status, pay, and self-regulation in return for valuable professional services.

Engineering fits this model well.

Business Model (Cynical View)

Societies act as **elite labour unions**
- limiting practitioners, controlling working conditions, and inflating salaries.

Engineering societies currently do NOT function this way.

What Is a Code of Ethics?

A code of ethics spells out the ways in which moral and ethical principles apply to professional practice

Framework

Provides a framework for ethical judgement of a professional

Decision Maker

Provides a starting point for ethical decision making - not a complete recipe

Not a Legal Document

Nobody can get arrested for violating a code of ethics

A code does **not** create new moral principles - it applies existing ones to professional practice. It should never be used as “window dressing.”

How a Code of Ethics Achieves Its Goals

Code of ethics helps in achieving workplace success

Creates an Environment

Helps create an environment where **ethical behaviour is the norm**

Bolsters Subordinate's Position

Provides **backup and support** against superiors who issue unethical orders

Shows Professionalism

The individual is viewed as a **strictly professional** person within the organisation

A code signals to the public that the profession is **seriously committed** to responsible conduct.

Objections to Codes of Ethics

If codes are so good, why doesn't every engineer follow one?

1. Not a Member

Engineers not in professional societies do not feel **compelled to abide** by the codes.

2. Lack of Awareness

Many practicing engineers are **unaware** that a code exists. Even members rarely consult it.

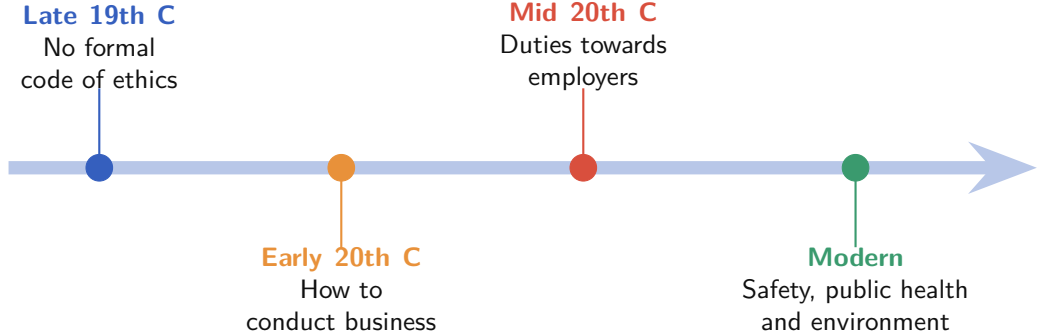
3. Internal Conflict

Codes can have **internal conflicts** with no guidance provided on how to resolve them.

Despite these objections, codes are in widespread use and generally serve a useful function.

Evolution of the Engineering Code of Ethics

How the code of ethics has developed over the years



The Two Codes of Ethics: IEEE vs. NSPE

	IEEE Code	NSPE Code
Length	Short; deals with generalities	Long and detailed
Coverage	Covers only the basics	Covers all the details
Readability	Easy to read and understand	Unlikely to be thoroughly read
Employer duty	Does <i>not</i> mention duty to employer	Section 1.4 - “faithful agents or trustees”
Environment	Explicitly mentions duty to protect the environment	No explicit mention

Trade-off: Shorter codes get read; longer codes cover more ground.

Resolving Internal Conflicts in Codes

Scenario: An employer orders an engineer to implement a design the engineer believes is **unsafe**.

1.NSPE CI. I.4

Engineers have a **duty to employer** - so should proceed with the design.

2.NSPE CI. I.1

Safety of the public is paramount - so should refuse the unsafe design.

3.Hierarchy

The code has a hierarchy. **Public safety always takes precedence.**

4.Generalising

Not all conflicts are easy. Hierarchy is not always explicitly stated.

Resolution: Engineer must refuse the unsafe design. Public safety comes first.

Can Codes and Societies Protect Employees?

The BART Case - A Real Example

- Three electrical engineers were **fired** from BART
- They reported **safety deficiencies** in the train control systems
- Cited the **IEEE code of ethics** - primary duty is public safety
- The **IEEE intervened in court** on their behalf
- Engineers ultimately **lost the case**

Key Lesson

Engineering societies are still **too weak** to fully protect engineers who report unethical behaviour.

For codes to have real meaning, this type of **intervention is essential**.

Since not all engineers are members, the pressure remains **very limited**.

Case Study 1: The Intel Pentium Chip

What Happened

- Flaw in the **floating-point unit**
- Intel knew about the issue **before public disclosure**
- Continued selling chips despite known defect
- **IBM** cancelled Pentium-based systems
- Total cost: **\$475 million** write-off

The Ethics Issues

- **Withholding known defects** - violates transparency
- **Misleading customers** - continued sales despite flaw
- **Loss of trust** in engineering accountability

Engineers are required to disclose known product risks honestly.

Case Study 2: Denver International Airport Runway Concrete

What Happened

- 3Bs Construction **diluted the concrete mix**
- Empire Laboratories **falsified test reports**
- Batch plant computers were **tampered with**
- Inspectors were **tipped off** in advance
- Investigated by FBI, grand juries, and Congress

The Ethics Issues

- **Falsifying data** - violates honesty and objectivity
- **Deceiving inspectors** - undermines safety oversight
- **Endangering the public** - runway structural risk

"Based upon engineering judgment" misused to justify altered data.

Case Study 3: The Paradyne Computer Case

What Happened

- SSA awarded a **\$115 million** contract
- Bid involved **P8400** - **not yet existing**
- Demonstration used a **relabelled DEC system**
- Acceptance tests failed; standards were **relaxed**
- Former SSA official involved in negotiation process

The Ethics Issues

- **Misrepresentation** - bidding on non-existent system
- **Deception** in procurement process
- **Conflict of interest** - revolving door employment

Agency oversight failures also contributed to the issue.

Key Takeaways

Chapter 2 - Summary

On Professionalism

- Engineering **is** a profession - but less mature than medicine and law
- Engineering societies are relatively **weak** compared to AMA and ABA
- Most engineers work as **employees** - not independent practitioners

On Codes of Ethics

- Codes are **frameworks** - not rulebooks or legal documents
- **Public safety always comes first** - overrides duty to employer
- Codes only work if engineers **read, know, and apply** them
- **No code replaces good judgment**

The three case studies show what happens when ethical standards are ignored - even at great cost.

Read the NSPE Codes of Ethics from
Engineering Ethics Concepts and Cases
(Charles E. Harris Jr)
— Appendix (Page 291)

Thank you